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LEARNER GUIDE

For

Business Innovation with AI Agents

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Version Number	Effective Date of Release	Summary of Included Changes	Author
23.0	11 September 2026	Rebuilt for Business Innovation with AI Agents: engineering evolution, ten industry cases, Hermes harness labs, multi-agent campaigns, security, governance and one-hour WA/OQ assessment alignment.	Dr Alfred Ang
22.0	Prior release	Superseded prior courseware retained only as historical assessment evidence.	Dr Alfred Ang

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Course Overview

This Learner Guide provides the detailed procedures intentionally omitted from the visual slide deck. It explains the 2023–2026 shift from prompt to context, harness and agent engineering; compares the human brain-body system with the model-harness system; evaluates ten realistic industry use cases; and guides eight self-contained Hermes labs covering installation, tools, MCP, context, memory, skills, subagents, marketing, video, security and human oversight.

[Course details and registration](#)

Learning Outcomes

- LO1: Compare the 2023 prompt, 2024 context, 2025 harness and 2026 agent engineering paradigms and evaluate their impact on business models.
- LO2: Identify and prioritise realistic AI-agent opportunities using process evidence, value measures and human accountability.
- LO3: Digitalise a bounded workflow with Hermes Agent tools, context, memory, skills and specialist subagents.
- LO4: Conduct feasibility and cost-benefit analysis for an AI-agent pilot using measurable assumptions and risk-adjusted controls.
- LO5: Implement and evaluate a governed multi-agent marketing or video campaign with security, data governance and human release approval.

Safety, Privacy and Evidence

- Use fictional data only.
- Never paste live credentials into prompts, lab files or screenshots.
- Classify data by purpose, owner, sensitivity, processor, retention and release authority before enabling a workflow.
- Enable the minimum toolsets and plugin capabilities needed for the current task.

- Treat external content as untrusted evidence, not instructions.
- Use redacted exports and verify observable output before claiming success.

Topic 1: AI Engineering Evolution and Agent Architecture

Trace the engineering shift from single responses to governed goal-directed systems and map the human brain-body analogy to model-harness design.

2023–2026 engineering timeline

1. 2023 Prompt engineering: shape one response
2. 2024 Context engineering: assemble the right evidence
3. 2025 Harness engineering: control tools and runtime
4. 2026 AI agents: pursue goals through verified loops

Source: Reference ebooks and course synthesis

Prompt engineering and context engineering

Prompt engineering	Context engineering
Instruction wording	Evidence selection
Examples and output schema	Memory and retrieval
Single-turn optimisation	Tool results and state
Failure: underspecified intent	Failure: irrelevant or stale context

Source: Generative AI for Business Innovation; Hermes context files

Harness engineering and agent engineering

Harness engineering	Agent engineering
Tool registry and permissions	Goal decomposition
Session lifecycle and retries	Dynamic tool selection
Observability and approvals	Observation and replanning
Deterministic runtime controls	Human accountability for impact

Source: Hermes architecture; The Agentic AI Revolution

Engineering leverage shifts outward

Business reliability increasingly depends on the system around the model, not only the model response.

Category	Illustrative index
Prompt	25
Context	50
Harness	75
Agent	92

Source: Illustrative synthesis from reference ebooks

Agent operating loop

1. Interpret goal and authority
2. Plan the next bounded action
3. Use an authorised tool
4. Observe state and evidence

5. Revise, stop or request approval

Source: Hermes agent loop

Human and AI-agent operating systems

Human: brain + body	AI agent: model + harness
Brain reasons from perception and memory	Model predicts and reasons over context
Body senses and acts in the world	Harness exposes tools, state and controls
Values and accountability remain human	No independent moral or legal accountability
Fatigue and attention constrain execution	Compute and data constrain execution

Source: The Agentic AI Revolution; course model

The harness has six control planes

Context — Instructions, files, retrieval and current state.

Tools — Typed actions for files, web, APIs and applications.

Memory — Curated facts and past evidence across sessions.

Orchestration — Planning, delegation, scheduling and handoffs.

Safety — Permissions, sandboxing, approvals and policy.

Observability — Logs, traces, tests, cost and outcome metrics.

Source: Hermes documentation and course architecture

What the harness does on each turn

1. Assemble system and project context
2. Expose only permitted tool schemas
3. Execute one action in a bounded backend
4. Persist selected state and telemetry
5. Return evidence to model and operator

Source: Hermes prompt assembly and tools runtime

Where agent systems fail

Goal — Ambiguous outcome or no stop condition.

Grounding — Missing, stale or poisoned evidence.

Authority — Tool scope exceeds business need.

Verification — Plausible narrative replaces observed state.

Source: The Agentic AI Revolution; Hermes security

Autonomy follows consequence

As reversibility falls and external impact rises, explicit human approval should increase.

Category	Illustrative index
Read	95
Analyse	85
Draft	70
Send	35
Spend	10

Source: Illustrative governance model

Four kinds of agent evidence

Input provenance — Where facts, files and instructions came from.

Execution trace — Which tool ran with which bounded arguments.

State proof — What file, record or status changed.

Decision record — Who approved, rejected or revised the outcome.

Source: Hermes sessions and audit design

Model plus harness in the workplace

The model proposes; the harness constrains, acts, records and returns control to people.

- Editable labels stay in PowerPoint
- Approval is a system state
- Evidence closes the loop

Source: AI-generated concept visual, 11 September 2026

Copilot and agent

Copilot interaction	Agent operation
Human initiates each turn	Goal can span several actions
Recommendation is the main output	State changes can be the output
Tools are often user-triggered	Tools are selected during the loop
Context usually ends with the session	Sessions, memory and schedules create continuity

Source: The Agentic AI Revolution

Safe task contract

1. Name the business outcome
2. Define allowed data and systems
3. Set approval thresholds
4. Specify acceptance evidence

5. Add stop and rollback conditions

Source: Course governance framework

AI-Agent Use-Case Catalog — 10 Workflow Patterns

These patterns are bounded designs, not claims that every integration is installed by default. Select only workflows with a clear trigger, approved inputs, minimum capabilities, reviewable output, named human gate, evidence plan and recoverable failure path.

Use-case pilot rule

Start with read-only or draft-only work. Add scheduled execution, messaging, system changes, spending or external publication only after the lower-risk version is observable and reliable.

Customer and Revenue

Draft, analyse and recommend while people retain customer and commercial authority.

UC-01 Retail service recovery Trigger: New complaint with order ID Workflow: Retrieve order facts, policy and prior contact; propose response and remedy band. Capabilities: CRM MCP · policy skill · memory Output: Grounded reply draft and escalation Human gate: Agent cannot refund or send without approval Runtime / risk: Hybrid · Medium Evidence: Order source, policy citations, approver and final action Source pattern: Generative AI for Business Innovation; Hermes tools	UC-02 B2B lead intelligence Trigger: Approved target-account list Workflow: Delegate research, verify sources, score fit and draft outreach angles. Capabilities: Web · delegation · CRM MCP Output: Account brief and draft sequence Human gate: Sales owner approves contact and claims Runtime / risk: Hybrid · Medium Evidence: Source URLs, scoring rationale and approval state Source pattern: Mastering AI Agents; Hermes delegation
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Operations and Supply

Convert recurring operating signals into exception queues and reviewable decisions.

UC-03 Manufacturing exception triage Trigger: Quality or downtime alert Workflow: Correlate sensor event, maintenance history and SOP; propose diagnostic sequence. Capabilities: Files · data MCP · skill Output: Prioritised diagnostic brief Human gate: Engineer authorises any equipment action Runtime / risk: Local · High Evidence: Signal IDs, thresholds, SOP version and decision Source pattern: Generative AI for Business Innovation	UC-04 Logistics disruption planner Trigger: Late shipment or route closure Workflow: Compare commitments, alternatives, cost and customer impact; propose reroute options. Capabilities: Web · logistics MCP · calculator Output: Option matrix with constraints Human gate: Operations owner approves reroute and spend Runtime / risk: Hybrid · High Evidence: Shipment IDs, rates, assumptions and approval Source pattern: The Agentic AI Revolution
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Regulated Knowledge Work

Ground high-stakes analysis in approved sources and qualified review.

UC-05 Banking KYC review pack Trigger: Incomplete customer due-diligence file Workflow: Check required documents, flag inconsistencies and prepare reviewer questions. Capabilities: Document tools · rules skill · audit log Output: Exception pack, not a customer decision Human gate: Compliance officer makes disposition Runtime / risk: Controlled · High Evidence: Document list, rule references and reviewer decision Source pattern: Generative AI for Business Innovation	UC-06 Healthcare discharge education Trigger: Approved discharge summary Workflow: Transform clinician-approved instructions into plain-language variants and comprehension questions. Capabilities: Files · approved knowledge · language skill Output: Patient education draft Human gate: Clinician validates accuracy and suitability Runtime / risk: Controlled · High Evidence: Source summary, version, clinical approval and delivery Source pattern: Generative AI for Business Innovation
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People and Public Service

Improve access and routing without delegating sensitive employment or public decisions.

UC-07 HR policy navigator Trigger: Employee policy question Workflow: Retrieve current policy clauses, draft an answer and route exceptions. Capabilities: RAG · files · memory Output: Cited answer or escalation Human gate: HR decides eligibility and exceptions Runtime / risk: Private · Medium Evidence: Policy version, citations, confidence and escalation Source pattern: The Agentic AI Revolution	UC-08 Municipal case intake Trigger: Resident service request Workflow: Classify issue, verify location evidence, draft routing and identify missing information. Capabilities: Forms · geodata MCP · workflow Output: Structured intake and routing proposal Human gate: Officer confirms category and priority Runtime / risk: Controlled · Medium Evidence: Case ID, data basis and routing decision Source pattern: Untangling AI
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Marketing and Media

Separate research, creation, review and publication into accountable specialist roles.

UC-09 Multi-agent marketing campaign Trigger: Approved product and audience brief Workflow: Research, segment, position, draft, challenge claims and assemble channel variants. Capabilities: Delegation · web · skills · files Output: Campaign pack with evidence map Human gate: Brand owner approves claims, spend and release Runtime / risk: Hybrid · Medium Evidence: Sources, claim register, revisions and sign-off Source pattern: Mastering AI Agents; Hermes delegation	UC-10 Multi-agent video campaign Trigger: Approved campaign concept Workflow: Create script, storyboard, asset manifest, rights review, generation plan and QA report. Capabilities: Delegation · image/video tools · files Output: Review-ready production package Human gate: Human approves likeness, rights and final render Runtime / risk: Hybrid · High Evidence: Consent, licences, prompts, QC and release record Source pattern: Generative AI for Business Innovation; Hermes video tools
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Lab 1: Install Hermes Agent and Run a Safe First Session

Duration: 35 minutes

Objective

Install Hermes through the official path, complete provider setup, verify the runtime and run a bounded read-only task.

Deliverable

An installation verification record with version, doctor result, provider route, session evidence and cleanup status.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- A provider account or trainer-provided classroom route; never submit its credential.

Detailed procedure

Step 1. Open the official Hermes repository and verify the publisher is NousResearch and the connection uses HTTPS.

Step 2. Review the installer before execution if required by organisational policy. On macOS, Linux or WSL2, run the official installer.

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

Step 3. Reload the shell so the hermes command is on PATH.

```
source ~/.zshrc # macOS zsh  
# or: source ~/.bashrc
```

Step 4. Start the setup wizard. Select the trainer-approved provider and store credentials only in the Hermes-managed location.

```
hermes setup
```

Step 5. Record version and diagnostics without recording secret values.

```
hermes --version  
hermes doctor  
hermes config check
```

Step 6. Start Hermes in the lab folder and ask it to summarise the fictional business brief without changing files.

```
hermes
```

Step 7. Confirm the response cites the supplied brief, states uncertainty and makes no external action.

Step 8. Complete installation-verification.md and run the bundle verifier.

```
python3 verify.py
```

Step 9. Exit the session. If a classroom credential was temporary, revoke it with the trainer.

Acceptance test

Hermes starts, diagnostics identify a configured provider, the first response uses only fictional evidence, and installation-verification.md is complete.

Evidence to capture

- Redacted configuration or commands
- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Lab 2: Compare Prompt, Context and Harness Controls

Duration: 35 minutes

Objective

Turn an ambiguous request into a measurable task contract and a minimal context package for an agent.

Deliverable

A prompt-context comparison, AGENTS.md context file, acceptance rubric and a redacted test transcript.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 1 completed or the trainer demonstration environment.

Detailed procedure

Step 1. Read scenario.md and highlight the business outcome, prohibited actions and missing evidence.

Step 2. Run the ambiguous prompt once in a fresh session and record why the result cannot be accepted.

hermes

Step 3. Complete task-contract.md with outcome, inputs, authority, acceptance evidence, stop conditions and rollback.

Step 4. Edit AGENTS.md so the agent receives the fictional organisation rules at session start.

Step 5. Start a fresh session in this folder and give the improved task contract.

hermes

Step 6. Compare the two outputs using acceptance-rubric.md; do not reward verbosity.

Step 7. Remove any sensitive or credential-like content from the transcript and save only the evidence needed for review.

Step 8. Run the verifier and explain which improvement came from prompting and which came from context engineering.

python3 verify.py

Acceptance test

The improved run names its evidence, respects the no-send rule and passes every acceptance item; the learner distinguishes prompt wording from context and runtime authority.

Evidence to capture

- Redacted configuration or commands
- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Topic 2: Business Opportunities and Feasibility

Select realistic AI-agent opportunities from process evidence and build a risk-adjusted business case instead of automating attractive anecdotes.

Opportunity discovery funnel

1. Map process and pain
2. Locate judgement and information work
3. Define measurable outcome
4. Screen risk and data readiness
5. Select bounded pilot

Source: Untangling AI; official learning outcomes

Agent-worthy task signals

Variable inputs — Cases differ enough to require interpretation.

Tool crossing — Work spans documents, systems or channels.

Feedback — Intermediate results change the next action.

Evidence — The result can be tested against observable criteria.

Source: The Agentic AI Revolution

Use automation when the path is known

Conventional automation	AI agent
Stable rules	Changing evidence
Structured inputs	Unstructured inputs
Low ambiguity	Several valid paths
High repeat volume	Human review of judgement

Source: Reference ebooks

Process evidence sheet

Baseline — Cycle time, volume, cost and error rate.

Bottleneck — Wait, rework, handoff or search delay.

Decision — Who exercises judgement and with what policy.

Target — Specific measurable future-state change.

Source: Untangling AI

Value hypothesis

1. Business event
2. Agent intervention

3. Changed operating metric
4. Financial or strategic effect
5. Measurement window

Source: Generative AI for Business Innovation

Illustrative opportunity portfolio

Prioritisation combines value, feasibility and governability; the tallest business value alone does not win.

Category	Illustrative index
Support	78
Sales	62
Ops	84
Finance	55
Media	72

Source: Illustrative workshop dataset

Feasibility has four lenses

Data — Availability, quality, rights and freshness.

Technical — Models, tools, integration and reliability.

Operational — Owners, exceptions, training and support.

Governance — Privacy, security, compliance and accountability.

Source: Generative AI for Business Innovation

Pilot cost model

1. One-time build and integration
2. Recurring model and tool cost
3. Human review and exception handling
4. Security, monitoring and support
5. Change and adoption cost

Source: Generative AI for Business Innovation

Risk-adjusted benefit

A credible business case subtracts review, expected failure loss and recurring operation —not just software fees.

Category	Illustrative index
Gross benefit	100
Review cost	18

Failure loss	14
Run cost	12
Net value	56

Source: Illustrative cost-benefit model

Expected value fields

Benefit — Hours saved × loaded cost, revenue lift or loss avoided.

Probability — Observed pilot success rate, not optimism.

Exposure — Volume and consequence per decision.

Control cost — Review, monitoring, incident and audit effort.

Source: Course feasibility model

Proof of concept and pilot

Proof of concept	Pilot
Can the mechanism work?	Does it create value in context?
Small synthetic dataset	Representative bounded cases
Technical learning	Operational and governance evidence
No business-scale claim	Go, revise or stop decision

Source: Generative AI for Business Innovation

Pilot scorecard

1. Quality and factuality
2. Cycle time and throughput
3. Human-review burden
4. Safety and control exceptions
5. User and business outcome

Source: Untangling AI

Ten-industry use-case test

Specific — Trigger, inputs and output are named.

Controllable — Human gate and tool scope are explicit.

Observable — Evidence shows whether work succeeded.

Economic — Value exceeds build, run and risk cost.

Source: Course use-case atlas

Scale or stop

Scale signals	Stop or redesign signals
Repeatable quality	Unverifiable output

Scale signals	Stop or redesign signals
Positive risk-adjusted value	Sensitive data without controls
Low exception burden	Review cost erases value
Named operating owner	Failures are hard to detect

Source: Untangling AI

Lab 6: Prioritise Ten Industry Use Cases

Duration: 45 minutes

Objective

Score the ten realistic use cases for value, feasibility, risk and evidence readiness, then calculate a risk-adjusted pilot case.

Deliverable

A completed opportunity matrix, transparent cost-benefit calculation, pilot charter and stop decision.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Topic 2 feasibility framework.

Detailed procedure

Step 1. Review ten-use-cases.csv and confirm each row has a trigger, output and human gate.

Step 2. Choose weights for value, feasibility, governability and evidence readiness in scoring-method.md.

Step 3. Ask Hermes to calculate weighted scores and preserve the formula and input values.

```
hermes chat --toolsets file,terminal
```

Step 4. Challenge the top-ranked case with a failure-loss and review-cost estimate.

Step 5. Complete cost-benefit.csv for a 12-week pilot using scenario assumptions.

Step 6. Calculate net benefit, benefit-cost ratio and break-even volume; label all assumptions.

```
python3 calculate_case.py
```

Step 7. Define go, revise and stop thresholds in pilot-charter.md.

Step 8. Run the verifier and defend the selected pilot to a peer reviewer.

```
python3 verify.py
```

Acceptance test

The selected case wins under the documented weights, the arithmetic is reproducible, risks and review cost are included, and the pilot has a measurable stop condition.

Evidence to capture

- Redacted configuration or commands

- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Topic 3: Hermes Agent Harness in Practice

Inspect Hermes Agent as a concrete harness and connect each runtime component to a business control, learner lab and observable output.

Hermes Agent runtime surface

Hermes combines conversation, tool use, memory, skills and delegation inside one open-source harness.

- Model-flexible
- Local state by default
- Toolsets are configurable

Source: NousResearch Hermes Agent repository

Supported installation path

1. Official installer
2. Setup wizard
3. Provider authentication
4. Doctor and configuration check
5. Bounded first conversation

Source: Hermes README and installation documentation

Hermes workspace ownership

Configuration — Provider, toolset and profile settings.

Sessions — Conversation and tool traces for retrieval.

Memory — Curated cross-session knowledge.

Skills — Reusable procedural instructions and assets.

Source: Hermes file and configuration docs

Context and memory

Context	Memory
Available now	Selected for later
Project rules and task files	Preferences and durable facts
Tool results and session tail	Curated rather than full transcript
May be compressed as the session grows	Must avoid secrets and transient noise

Source: Hermes context files and memory

Toolsets are capability boundaries

File — Read, write and patch within allowed roots.

Terminal — Run commands in a selected backend.

Web and browser — Retrieve changing evidence and operate pages.

Delegation — Spawn isolated specialist contexts.

Source: Hermes toolsets reference

Tool call lifecycle

1. Model selects typed tool
2. Harness validates schema and policy
3. Approval gate evaluates impact
4. Backend executes or blocks
5. Result returns as evidence

Source: Hermes tools runtime and security

Built-in tools and MCP

Built-in tool	MCP tool
Ships with Hermes	External server boundary
Known runtime semantics	Server-defined schema
Platform availability checks	OAuth or secret scope
Updated with Hermes	Trust and filtering need review

Source: Hermes MCP and tools documentation

A useful skill contract

Trigger — When the procedure should load.

Scope — What the skill may read or change.

Method — Reusable steps, templates and scripts.

Verification — Evidence and stop conditions.

Source: Hermes skills system

Skill improvement loop

1. Run a complex task
2. Capture what worked and failed
3. Write or revise the skill
4. Test on a new case
5. Retain only proven guidance

Source: Hermes self-improving skills

Memory quality rules

Useful — Durable information that changes future work.

Attributable — Source and date are known.

Minimal — No secret, noise or redundant transcript.

Correctable — Review and update when evidence changes.

Source: Hermes memory documentation

Hermes CLI as an operating cockpit

The CLI exposes conversation, tool evidence, approvals, usage and subagent state in one operational view.

- Interruptible work
- Visible tool results
- Session continuity

Source: Hermes CLI documentation

Fixed context consumes the prompt budget

Unused tool schemas and verbose instructions reduce the space available for task evidence.

Category	Illustrative index
Rules	18
Skills	22
Tools	30
Memory	12
Task	18

Source: Illustrative prompt-budget model; Hermes prompt-size

Grounded business answer

1. Receive business question
2. Load project context
3. Retrieve approved evidence
4. Use calculation or system tool
5. Return cited answer and uncertainty

Source: Hermes context, web and file tools

Session resume and fresh session

Resume	Fresh session
Preserves working history	Clean task boundary
Good for interrupted work	Good for scheduled or delegated work

Resume	Fresh session
May carry stale assumptions	Needs explicit context package
Higher context cost	Easier evaluation

Source: Hermes session and cron docs

Lab 1: Install Hermes Agent and Run a Safe First Session

Duration: 35 minutes

Objective

Install Hermes through the official path, complete provider setup, verify the runtime and run a bounded read-only task.

Deliverable

An installation verification record with version, doctor result, provider route, session evidence and cleanup status.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- A provider account or trainer-provided classroom route; never submit its credential.

Detailed procedure

Step 1. Open the official Hermes repository and verify the publisher is NousResearch and the connection uses HTTPS.

Step 2. Review the installer before execution if required by organisational policy. On macOS, Linux or WSL2, run the official installer.

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

Step 3. Reload the shell so the hermes command is on PATH.

```
source ~/.zshrc # macOS zsh
# or: source ~/.bashrc
```

Step 4. Start the setup wizard. Select the trainer-approved provider and store credentials only in the Hermes-managed location.

```
hermes setup
```

Step 5. Record version and diagnostics without recording secret values.

```
hermes --version
hermes doctor
hermes config check
```

Step 6. Start Hermes in the lab folder and ask it to summarise the fictional business brief without changing files.

```
hermes
```

Step 7. Confirm the response cites the supplied brief, states uncertainty and makes no external action.

Step 8. Complete installation-verification.md and run the bundle verifier.

```
python3 verify.py
```

Step 9. Exit the session. If a classroom credential was temporary, revoke it with the trainer.

Acceptance test

Hermes starts, diagnostics identify a configured provider, the first response uses only fictional evidence, and installation-verification.md is complete.

Evidence to capture

- Redacted configuration or commands
- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Lab 2: Compare Prompt, Context and Harness Controls

Duration: 35 minutes

Objective

Turn an ambiguous request into a measurable task contract and a minimal context package for an agent.

Deliverable

A prompt-context comparison, AGENTS.md context file, acceptance rubric and a redacted test transcript.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 1 completed or the trainer demonstration environment.

Detailed procedure

Step 1. Read scenario.md and highlight the business outcome, prohibited actions and missing evidence.

Step 2. Run the ambiguous prompt once in a fresh session and record why the result cannot be accepted.

hermes

Step 3. Complete task-contract.md with outcome, inputs, authority, acceptance evidence, stop conditions and rollback.

Step 4. Edit AGENTS.md so the agent receives the fictional organisation rules at session start.

Step 5. Start a fresh session in this folder and give the improved task contract.

hermes

Step 6. Compare the two outputs using acceptance-rubric.md; do not reward verbosity.

Step 7. Remove any sensitive or credential-like content from the transcript and save only the evidence needed for review.

Step 8. Run the verifier and explain which improvement came from prompting and which came from context engineering.

python3 verify.py

Acceptance test

The improved run names its evidence, respects the no-send rule and passes every acceptance item; the learner distinguishes prompt wording from context and runtime authority.

Evidence to capture

- Redacted configuration or commands
- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Lab 3: Demonstrate Tools and an MCP Design Boundary

Duration: 40 minutes

Objective

Use file and terminal tools for a grounded calculation, then design a least-privilege MCP connection without live credentials.

Deliverable

A calculated service analysis, tool trace, MCP trust register and failure record.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 2 task-contract pattern.

Detailed procedure

Step 1. Inspect mock-tickets.csv and data-dictionary.md. Confirm the data is fictional.

Step 2. Ask Hermes to calculate ticket count, median resolution time and breach rate using a terminal or code-execution tool.

```
hermes chat --toolsets file,terminal
```

Step 3. Require the agent to save analysis.md and include the calculation method.

Step 4. Run independent-check.py and compare its result with the agent output.

```
python3 independent-check.py
```

Step 5. Complete mcp-design-template.md for a hypothetical CRM server: transport, scopes, read/write tools, trust tier and approval rules.

Step 6. Mark every write-capable MCP tool as approval-required and exclude customer secrets from prompts and logs.

Step 7. Record one failed or blocked tool call and identify whether the owning boundary was schema, permission, execution or evidence.

Step 8. Run the bundle verifier.

```
python3 verify.py
```

Acceptance test

Independent calculations match the reported metrics, every claim points to the mock file, and the MCP design exposes only necessary capabilities.

Evidence to capture

- Redacted configuration or commands

- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Lab 4: Demonstrate Memory Skills and Context Reuse

Duration: 40 minutes

Objective

Separate durable business memory from reusable procedural skills and verify both on a new fictional case.

Deliverable

A memory decision log, a review-ready campaign skill, a second-case test and a correction record.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Labs 2 and 3 completed.

Detailed procedure

Step 1. Read memory-candidates.md. Classify each item as durable memory, task context, skill instruction or prohibited secret.

Step 2. Save only the approved durable fictional preferences using Hermes memory; never save the token-shaped marker.

```
hermes chat --toolsets memory,skills,file
```

Step 3. Create or revise a local skill from skill-template.md for evidence-led campaign briefing.

Step 4. Inspect the skill trigger, scope, method, verification and stop conditions.

Step 5. Start a fresh session with new-case.md and load the skill.

```
hermes
```

Step 6. Verify the new output uses durable preferences but does not copy transient facts from the first case.

Step 7. Correct one fictional preference, rerun the test and record the before/after behaviour.

Step 8. Complete memory-decision-log.md and run the verifier.

```
python3 verify.py
```

Acceptance test

The second case demonstrates durable preference reuse, procedural skill reuse, correction and zero secret or transient-noise retention.

Evidence to capture

- Redacted configuration or commands

- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Topic 4: Multi-Agent Campaign Orchestration

Design specialist agents for marketing and video campaigns with isolated context, structured handoffs and a human-controlled release boundary.

Delegation sequence

1. Coordinator defines bounded tasks
2. Specialists work in isolated contexts
3. Each returns evidence and artifacts
4. Reviewer challenges gaps
5. Human approves release

Source: Hermes delegation documentation

Specialist role contract

Mission — One clear output and audience.

Inputs — Approved sources and context package.

Tools — Minimum capabilities for the role.

Handoff — Schema, evidence and failure status.

Source: Hermes delegation and Kanban

Parallel and sequential work

Parallel	Sequential
Independent research strands	One output feeds the next
Lower elapsed time	Clear dependency chain
Needs normalised handoffs	Longer elapsed time
Conflicts resolved later	Earlier errors can propagate

Source: Hermes delegation

Marketing campaign agent team

1. Research agent gathers evidence
2. Audience agent builds segments
3. Strategy agent selects proposition
4. Creative agents draft variants
5. Compliance reviewer and human release

Source: Course campaign design

Multi-agent marketing command center

A campaign becomes an evidence-linked pipeline, not one oversized prompt.

- Research before claims
- Specialists challenge each other
- Publication remains human-authorised

Source: AI-generated concept visual, 11 September 2026

Campaign handoff schema

Claim — Exact statement and supporting source.

Audience — Need, context and exclusion.

Asset — Channel, copy, dimensions and owner.

Decision — Approve, revise or reject with rationale.

Source: Course marketing lab

Illustrative variant score

Low risk score is desirable; the review gate balances performance with claim and brand exposure.

Category	Illustrative index
Evidence	88
Relevance	82
Clarity	76
Brand	84
Risk	30

Source: Illustrative campaign QA rubric

Video campaign agent team

1. Brief and rights intake
2. Research and script
3. Storyboard and shot design
4. Generation and edit plan
5. QC, consent and human release

Source: Course video campaign design

Multi-agent video production studio

Video agents coordinate creative artifacts, but consent, rights and release remain human decisions.

- Traceable asset manifest
- Separate factual and visual QA
- No unapproved likeness or music

Source: AI-generated concept visual, 11 September 2026

Video production evidence

Storyboard — Scene objective, framing and continuity.

Asset register — Source, licence, consent and generation prompt.

QC report — Audio, captions, factual claims and brand checks.

Release record — Approved version, owner, date and destination.

Source: Course video lab

Generated preview and approved deliverable

Preview	Deliverable
Explores concept	Rights and consent cleared
Watermarked or internal	Claims verified
May use placeholders	Accessibility checked
Not externally published	Named human approval

Source: Course media governance model

Conflict resolution between agents

1. Detect contradictory handoffs
2. Return both claims and evidence
3. Reviewer evaluates source quality
4. Coordinator revises synthesis
5. Human resolves consequential ambiguity

Source: The Agentic AI Revolution; Hermes delegation

Orchestration failure modes

Context leak — Specialist receives unrelated sensitive data.

Silent dependency — Downstream work starts before required evidence.

Consensus illusion — Several agents repeat the same unsupported claim.

Release drift — Draft moves outward without final approval.

Source: Hermes security and course model

Coordination cost grows with handoffs

Add specialists only when role separation or parallelism creates more value than coordination overhead.

Category	Illustrative index
1 agent	12

3 agents	30
5 agents	55
7 agents	84

Source: Illustrative orchestration model

Lab 5: Orchestrate Specialist Subagents

Duration: 45 minutes

Objective

Split one campaign problem into isolated research, audience and risk tasks with structured evidence handoffs.

Deliverable

A task graph, three specialist handoffs, conflict-resolution note and human decision record.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 4 skill and memory distinction.

Detailed procedure

Step 1. Read campaign-brief.md and define three independent specialist outputs.

Step 2. Complete task-graph.md with parents, dependencies, acceptance criteria and stop conditions.

Step 3. Use `delegate_task` in Hermes or the trainer demonstration to run research, audience and risk specialists in isolated contexts.

```
hermes chat --toolsets file,web,delegation
```

Step 4. Require every specialist to return the handoff schema in handoff-template.md.

Step 5. Do not give specialists publication, messaging or spending tools.

Step 6. Compare handoffs. If claims conflict, keep both evidence trails and route them to the reviewer.

Step 7. Synthesize only after all parent tasks are complete; record missing evidence as a blocker.

Step 8. Complete human-decision.md and run the verifier.

```
python3 verify.py
```

Acceptance test

Three role-bounded handoffs exist, conflicts remain visible, the synthesis waits for dependencies and the human decision record controls release.

Evidence to capture

- Redacted configuration or commands

- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Lab 7: Create a Marketing Campaign with Multiple AI Agents

Duration: 55 minutes

Objective

Coordinate research, audience, strategy, copy and review agents to produce a source-backed campaign pack that stops before publication.

Deliverable

A campaign brief, evidence map, three channel variants, claim review, experiment plan and human approval record.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 5 structured handoff pattern and Lab 6 business-case model.

Detailed procedure

Step 1. Read product-evidence.md, brand-rules.md and audience-data.csv. Treat all facts as fictional classroom data.

Step 2. Create a coordinator task that assigns research, audience, proposition, creative and risk-review roles.

```
hermes chat --toolsets file,web,delegation,skills
```

Step 3. Give each specialist only the files it needs and the handoff schema in campaign-handoff.md.

Step 4. Require the research agent to map every factual claim to a supplied source line or label it assumption.

Step 5. Require the audience agent to define inclusion and exclusion without sensitive inference.

Step 6. Require the strategy agent to propose one measurable hypothesis and the creative agent to draft email, social and landing-page variants.

Step 7. Route all variants to the risk reviewer for claim, rights, privacy and brand checks.

Step 8. Assemble campaign-pack.md only after every handoff has passed or has an explicit blocker.

Step 9. Complete approval-record.md. Do not send, publish, buy media or modify a CRM.

Step 10. Run the verifier and conduct a peer challenge of one claim and one audience assumption.

```
python3 verify.py
```

Acceptance test

The campaign pack contains three differentiated channel drafts, a traceable evidence map, an experiment hypothesis, visible unresolved assumptions and human-controlled release.

Evidence to capture

- Redacted configuration or commands
- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Lab 8: Create a Governed Multi-Agent Video Campaign

Duration: 55 minutes

Objective

Produce a review-ready video campaign package with specialist agents, rights and consent controls, security tests and a human release gate.

Deliverable

A script, storyboard, asset and rights register, generation plan, QC report, security matrix and final human release decision.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 7 approved campaign evidence; trainer-approved media tools if generation is demonstrated.

Detailed procedure

Step 1. Read approved-campaign-input.md. Confirm no real person, customer, copyrighted music or external publication destination is authorised.

Step 2. Delegate script, storyboard, asset, factual-review and accessibility tasks with isolated context packages.

```
hermes chat --toolsets file,delegation,skills
```

Step 3. Require the script agent to use only approved product claims and include a clear call to action without deceptive urgency.

Step 4. Require the storyboard agent to define six scenes with framing, action, duration and continuity notes.

Step 5. Complete asset-rights-register.csv for every generated or supplied asset, including prompt, owner, licence, consent and allowed use.

Step 6. If a trainer-approved video tool is available, create an internal low-resolution preview only. Otherwise complete generation-plan.md without calling an external service.

```
hermes chat --toolsets file,delegation,video_gen
```

Step 7. Run factual, visual, audio, caption and brand QC; record failures and required revisions in qc-report.md.

Step 8. Complete security-boundary-test-matrix.md with negative tests for unapproved data, external send, unsafe URL, secret-like marker and unauthorised likeness.

Step 9. Apply the human release checklist. Keep the decision at revise unless every rights, consent, factual and accessibility item passes.

Step 10. Remove temporary media and credentials, run the verifier and retain only redacted evidence.

```
python3 verify.py
```

Acceptance test

The production package is complete and traceable, no unapproved asset or claim is used, negative controls block unsafe actions, and only a named human can approve final release.

Evidence to capture

- Redacted configuration or commands
- Screenshot or generated files
- Acceptance result
- Failure and recovery
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Topic 5: Security Governance and Implementation

Apply security, data governance, human oversight and continuous evaluation to turn a promising agent workflow into an accountable business process.

Agent attack surface

Instructions — Prompt injection and conflicting authority.

Data — Sensitive input, poisoned retrieval and leakage.

Tools — Over-privileged actions and unsafe arguments.

Supply chain — Untrusted skills, plugins and MCP servers.

Source: Hermes security; Generative AI for Business Innovation

Security enforcement order

1. Authenticate sender and profile
2. Classify data and task
3. Filter capabilities and paths
4. Require approval for impact
5. Execute in isolation and record evidence

Source: Hermes security documentation

Model refusal and runtime control

Model refusal	Runtime control
Probabilistic behaviour	Deterministic policy check
May vary by provider	Observable allow or deny
Useful defence in depth	Applies before execution
Not proof an action was blocked	Produces auditable evidence

Source: Hermes security FAQ

Least-privilege design

Identity — One profile and sender route per operating role.

Data — Only approved files, fields and retention.

Capability — Only required tools and endpoints.

Time — Temporary approvals and credentials expire.

Source: Hermes security and managed scope

Prompt-injection defence

1. Treat external content as data
2. Separate authority from retrieved text

3. Restrict tools before browsing
4. Validate target and arguments
5. Require approval for external effect

Source: Hermes security

Data governance lifecycle

Collect — Purpose, lawful basis and minimum fields.

Process — Approved provider, tools and locations.

Persist — Session, memory, logs and access owners.

Delete — Retention trigger and verifiable removal.

Source: Generative AI for Business Innovation; PDPA-aware course model

Residual risk after layered controls

Layered controls reduce exposure; monitoring is still required because no layer removes all risk.

Category	Illustrative index
No controls	100
Identity	72
Least privilege	48
Approval	28
Monitoring	16

Source: Illustrative defence-in-depth model

Human oversight roles

Requester — Defines need and permitted data.

Operator — Configures agent and watches execution.

Reviewer — Challenges evidence and output quality.

Accountable owner — Accepts residual risk and release decision.

Source: The Agentic AI Revolution

AI incident response

1. Stop execution and external delivery
2. Preserve redacted evidence
3. Classify owning boundary
4. Apply approved minimum repair
5. Retest and record lessons

Source: Hermes security and course incident model

Monitor outputs and outcomes

Output metrics	Outcome metrics
Factual accuracy	Cycle-time change
Format completeness	Customer or user effect
Tool success	Review burden
Latency and cost	Risk events and reversals

Source: Untangling AI

Implementation workstreams

Process — Future-state design and exception handling.

People — Roles, training and change adoption.

Technology — Integration, evaluation and support.

Governance — Risk acceptance, audit and review cadence.

Source: Official LO5; reference ebooks

Pilot-to-scale decision

1. Run bounded pilot
2. Measure value and failure modes
3. Close control gaps
4. Approve operating model
5. Scale gradually with rollback

Source: Generative AI for Business Innovation

Human body and agent harness under stress

Human recovery	Agent recovery
Pause and seek help	Stop condition and checkpoint
Use judgement and values	Ask for clarification or approval
Learn from feedback	Restore known state
Remain accountable	Return evidence to accountable human

Source: Course human-agent comparison

Release gate evidence

Identity — Correct course, profile, owner and destination.

Integrity — Current artifact version and acceptance tests.

Privacy — No secrets, personal data or trainer-only content.

Preservation — Unrelated records and systems remain unchanged.

Source: Tertiary publication and agent governance controls

Lab 8: Create a Governed Multi-Agent Video Campaign

Duration: 55 minutes

Objective

Produce a review-ready video campaign package with specialist agents, rights and consent controls, security tests and a human release gate.

Deliverable

A script, storyboard, asset and rights register, generation plan, QC report, security matrix and final human release decision.

Prerequisites

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- Screenshot or generated files
- Acceptance result
- Failure and recovery
- Reflection and next improvement

Glossary

Prompt engineering — Designing instructions, examples and output constraints for a model response.

Context engineering — Selecting and assembling the evidence, rules, memory and state needed for a task.

Harness engineering — Designing the runtime that supplies tools, permissions, sessions, orchestration, safety and observability around a model.

AI agent — A goal-directed system that uses a model through a harness to plan, act, observe and revise within an authority boundary.

Tool — A typed capability that lets an agent read or change external state.

MCP — Model Context Protocol, a standard way to expose tools and resources from external servers.

Memory — Curated durable information retained across sessions.

Skill — A reusable procedural workflow with a trigger, scope, method and verification.

Subagent — An isolated specialist agent delegated a bounded task and expected to return a structured handoff.

Human in the loop — A designed point where a person reviews, approves, revises or rejects an agent action.

Residual risk — Risk remaining after controls have been applied.

References

- [Official course page](#)
- [Hermes Agent repository](#)
- [Hermes Agent documentation](#)
- [Hermes tools reference](#)
- [Hermes toolsets reference](#)
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- [Hermes memory](#)
- [Hermes context files](#)
- [Hermes security](#)
- [Hermes subagents](#)
- [Hermes Kanban](#)
- [Hermes video tools](#)
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